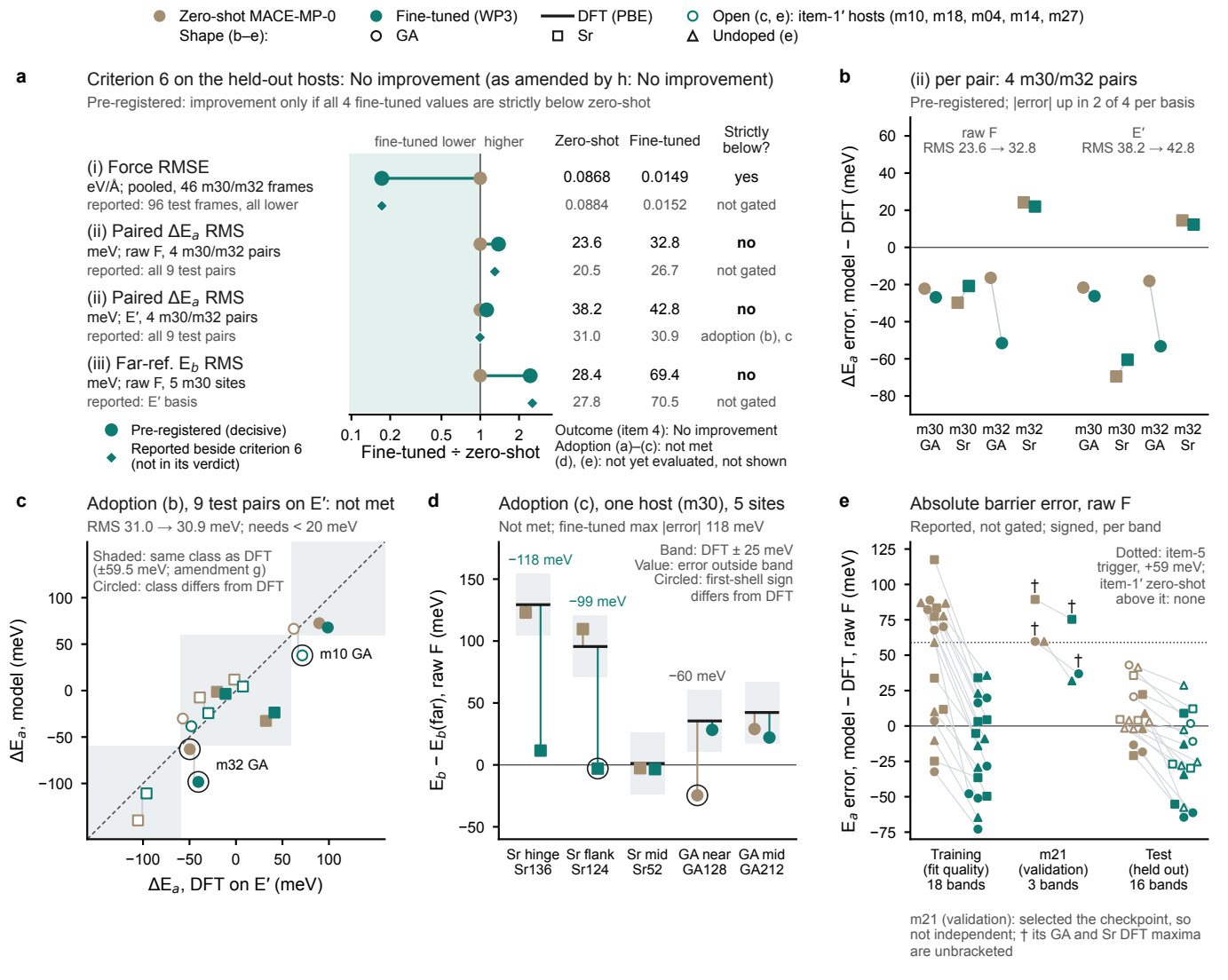

Zero-shot MACE-MP-0
Shape (b–e):
Fine-tuned (WP3)
GA
DFT (PBE)
Sr
Open (c, e): item-1′ hosts (m10, m18, m04, m14, m27)
Undoped (e)
a
Criterion 6 on the held-out hosts: No improvement (as amended by h: No improvement)
Pre-registered: improvement only if all 4 fine-tuned values are strictly below zero-shot
fine-tuned lower higher
Zero-shot
Fine-tuned
Strictly below?
(i) Force RMSE
eV/Å; pooled, 46 m30/m32 frames
reported: 96 test frames, all lower
0.0868
0.0149
yes
0.0884
0.0152
not gated
(ii) Paired ΔE$_a$ RMS
meV; raw F, 4 m30/m32 pairs
reported: all 9 test pairs
23.6
32.8
no
20.5
26.7
not gated
(ii) Paired ΔE$_a$ RMS
meV; E′, 4 m30/m32 pairs
reported: all 9 test pairs
38.2
42.8
no
31.0
30.9
adoption (b), c
(iii) Far-ref. E$_b$ RMS
meV; raw F, 5 m30 sites
reported: E′ basis
28.4
69.4
no
27.8
70.5
not gated
Pre-registered (decisive)
Reported beside criterion 6 (not in its verdict)
0.1 0.2 0.5 1 2
Fine-tuned ÷ zero-shot
Outcome (item 4): No improvement
Adoption (a)–(c): not met
(d), (e): not yet evaluated, not shown
b
(ii) per pair: 4 m30/m32 pairs
Pre-registered; |error| up in 2 of 4 per basis
raw F
RMS 23.6 → 32.8
E′
RMS 38.2 → 42.8
ΔE$_a$ error, model − DFT (meV)
60 40 20 0 −20 −40 −60 −80
m30 GA
m30 Sr
m32 GA
m32 Sr
m30 GA
m30 Sr
m32 GA
m32 Sr
c
Adoption (b), 9 test pairs on E′: not met
RMS 31.0 → 30.9 meV; needs < 20 meV
Shaded: same class as DFT (±59.5 meV; amendment g)
Circled: class differs from DFT
m10 GA
m32 GA
ΔE$_a$, model (meV)
100 50 0 −50 −100
−100 −50 0 50 100
ΔE$_a$, DFT on E′ (meV)
d
Adoption (c), one host (m30), 5 sites
Not met; fine-tuned max |error| 118 meV
Band: DFT ± 25 meV
Value: error outside band
Circled: first-shell sign differs from DFT
−118 meV
−99 meV
−60 meV
E$_b$ − E$_b$(far), raw F (meV)
150 100 50 0 −50
Sr hinge Sr136
Sr flank Sr124
Sr mid Sr52
GA near GA128
GA mid GA212
e
Absolute barrier error, raw F
Reported, not gated; signed, per band
Dotted: item-5 trigger, +59 meV; item-1′ zero-shot above it: none
E$_a$ error, model − DFT, raw F (meV)
125 100 75 50 25 0 −25 −50 −75
Training (fit quality) 18 bands
m21 (validation) 3 bands
Test (held out) 16 bands
m21 (validation): selected the checkpoint, so not independent; † its GA and Sr DFT maxima are unbracketed